Flight Mechanics and Aerodynamics of the MERA Windex

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Revisions

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5.0	Jan 13, 2023	Updated version for 2023	U. Ringertz (UR)
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1 Introduction

Electric propulsion for aircraft is gaining interest and momentum but the design of efficient such aircraft is difficult because of the fairly low energy density of current generation batteries. Efficiency can be improved by optimizing the design so that the aircraft approaches the aerodynamic efficiency of sailplanes (gliders) which have low drag to lift ratios and thus requires modest propulsion power.

The project work for the Flight Mechanics course is to develop a simulation model for the More Electric Research Aircraft (MERA) which is shown in Figure 1. The MERA is based on the Windex motor glider

Figure 1: The MERA Windex in flight.

which was designed and developed by the late KTH Professor Sven-Olof Ridder and co-workers. A history and background of the aircraft development is available in the reference documentation [1].

The electric version in the form of MERA was developed by Saab under a contract with the Swedish Defense Materiel Administration (FMV). A brief description of the development process is given in [6].

Your task is to develop a flight dynamics simulation model of the aircraft and then to perform stability and control analysis using the simulation model. The aerodynamic data for the simulation model will be

developed using a nonlinear lifting line model [3] in combination with airfoil analysis using Xfoil [2]. The aerodynamics analysis models will be compared to and enhanced using wind tunnel test data.

Your project assignments are listed in the different sections and your complete analysis should be presented in one report developed in stages as the course continues. There additional advanced topics indicated by * for those aiming for the higher grades.

1.1 Basic design data

The planform of the aircraft is shown in Figure 2 illustrating the basic characteristics with a straight tapered wing and a T-tail empennage. Reference geometric properties are listed in Table 1.

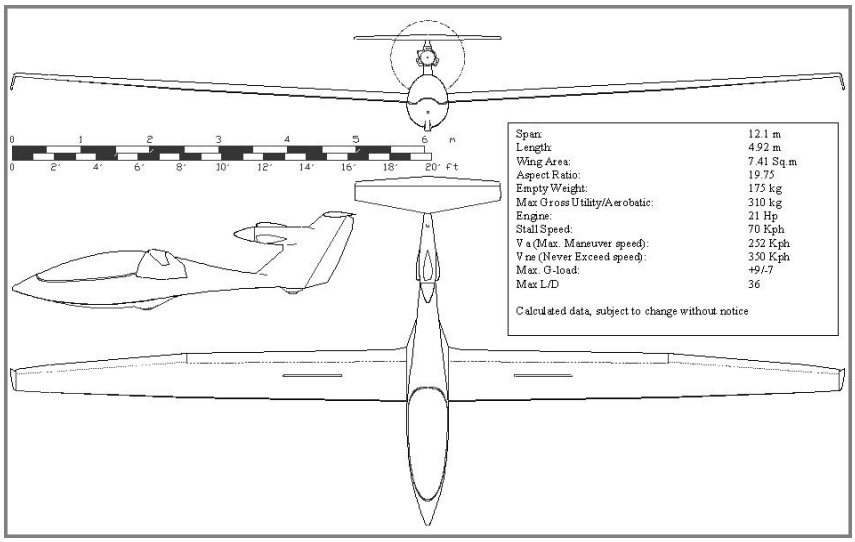

Figure 2: The Windex planform.

The mass properties for the MERA Windex are listed in Table 2. The reference datum for mass properties is the wing leading edge at the wing root. The orientation of the aircraft when mass properties are measured should be such that the fin leading edge is at 40 degrees to the level plane. This data is for the aircraft with propulsion system and batteries installed. You should add you personal body mass with typical summer clothing, parachute and optional ballast. If your body mass is low, you may need to add lead ballast in the nose of the aircraft.

There is CAD models available for download from the course page on canvas. There are two STL files of different resolution and also a STEP file which shown in Figure 3. The STL files can be viewed in

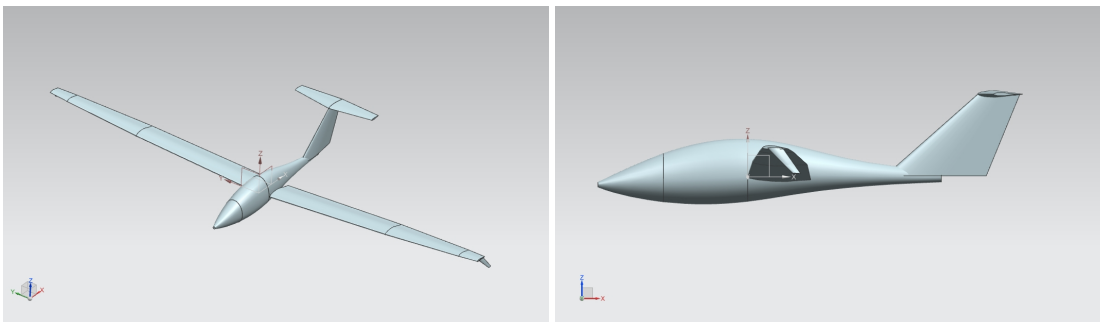

Figure 3: The Windex CAD model in STEP format.

Matlab using the script provided on canvas. Using the STEP model requires access to a CAD system. Note

		Full scale	Model 1:5
Reference span	b	12.1 m	2.42 m
Reference chord	c	0.633 m	0.1266 m
Reference wing area	S	7.41 m^2	0.2964 m^2
Aspect ratio	$Æ$	19.76	
Length		4.92 m	
Aileron to chord ratio	E	0.225	
Aileron differential	δ_a	+15 to -15 (deg)	
Horizontal tail			
Span	b_h	2.18 m	
Area	S_h	0.87 m^2	
Aspect ratio	$Æ_h$	5.4	
Elevator to chord ratio	E_h	0.3	
Elevator travel	δ_e	$\pm$ 30 (deg)	
Vertical tail			
Span	b_v	1.05 m	
Area	S_v	0.95 m^2	
Aspect ratio	$Æ_v$	1.28	
Rudder to chord ratio	E_v	0.28	
Rudder travel	δ_r	$\pm$ 30 (deg)	

Table 1: Reference data.

Empty mass	190 kg
Center of mass (behind datum)	0.507 m
c.g pilot	-0.340 m
Parachute	5 kg
c.g ballast	-1.670 m
I_{xx} roll-axis	700 kgm^2
I_{yy} pitch-axis	215 kgm^2
I_{zz} yaw-axis	900 kgm^2

Table 2: Inertial properties.

that the CAD model is in development and only partially correct in detail. Revised versions will be made available on canvas as they are developed.

Task 1: Preliminary analysis

Perform a preliminary analysis of the Windex aircraft based on the geometry data provided on canvas and suitable calculations for simplified aerodynamic data such as available by Ridder [7]. Estimate the following properties:

1. Estimate the neutral point
2. Estimate a suitable range of allowed locations of the center of mass
3. Determine if additional ballast is needed for you to fly
4. Compute the vertical tail volume ratio
5. Compute the horizontal tail volume ratio

Start a draft edition of your project report and write a brief introduction about the aircraft and your preliminary results.

Task 2: Nonlinear lifting line modeling

The purpose of this task is to become familiar with the nonlinear lifting line model for analysis of aircraft aerodynamics. The computational model is to be developed using the Matlab package made available on canvas in combination with airfoil data obtained using xfoil. Wind tunnel test data of a finite wing model is available for validation and to illustrate both the power of the modeling but also some limitations.

Numerical model

Create a computational lifting line model for the trapezoidal wing geometry used in the second lab which is shown in Figure 4. The reference data is given in Table 3. There are further drawings and complete CAD geometry available on canvas.

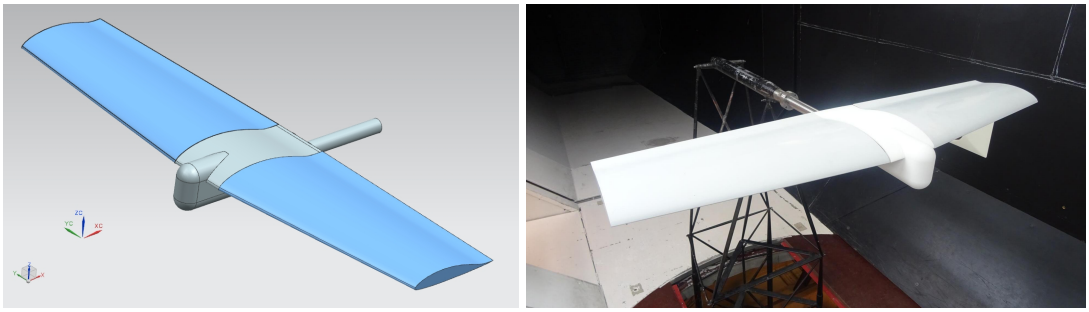

Figure 4: Wind tunnel model of the finite wing.

Reference area	S_{ref}	0.2781	m^2
Reference span	b_{ref}	1.125	m
Reference chord	c_{ref}	0.2471	m

Table 3: Reference data for the finite wing.

Use the provided airfoil data for the airfoil named H15-5 to predict lift, drag and pitch moment characteristics and neutral point of the airfoil. There is also a similar airfoil named H16-5 provided that is optimized for the inner part of a wing panel. The H15-5 airfoil is designed for use close to the wing tip of a glider. These airfoil data can then be used in the nonlinear lifting line analysis to estimate the corresponding data for the finite wing with this airfoil. You can chose to use xflr5 or the provided matlab package for the analysis or develop your own analysis.

Presentation of results

In your report, please provide curves of lift and moment as a function of angle of attack from both the analysis and the experiment. There should also be a drag polar showing the drag coefficient for different values of the lift coefficient.

Task 3: Steady aerodynamics data

Use the geometry data and drawings available on canvas to develop a nonlinear lifting line model of the Windex configuration. Wind tunnel test data is available for estimating the influence of the fuselage. Sample code for defining the geometry and aerodynamics is available on canvas in the folder `nonlinear_lifting_line/windex_matlab`.

Use your analysis model to estimate all steady state aerodynamic forces in nondimensional coefficient form. You will need to decide which force and moment coefficients are to be assumed constant and which will need to be nonlinear functions implemented using tabular lookup or approximated using spline functions.

Use your aerodynamics model to improve your estimate of:

1. the neutral point,
2. a suitable range of allowed locations of the center of mass,
3. needed ballast for you to fly the Windex.

Further, make sure to perform the following analysis:

1. Plot untrimmed C_L vs α and trimmed $C_{L\text{trim}}$ vs α_{trim}
2. Plot elevator to trim vs airspeed for your range of allowed center of mass (CoM).
3. One curve for your favourite CoM position, one for max forward, and one for max rear.
4. Compute a trimmed glide polar with sink rate on the vertical axis and flight speed on the horizontal axis. There are numerous examples of glider polars on the Internet to compare with.

Advanced topics*

Implement additional airfoil data for flap and aileron control. Use the drawings on canvas to estimate range of control surface settings and appropriate Reynolds number.

Extend the provided Matlab functions to also facilitate flap and aileron controls.

Wind tunnel test data

A wind tunnel model of the fuselage with chopped wings has been designed and manufactured. The model features a straight short wing with total model span of 0.6 meters and the model scale is 1:5. The model is shown in Figure 5. The STL CAD model is available in the `windex_cad` folder.

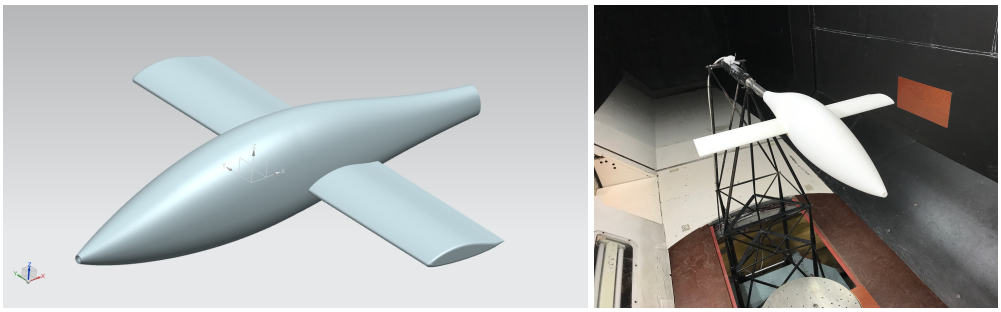

Figure 5: Wind tunnel model of fuselage section.

In order to separately estimate the influence of the short wing, there is also NLL Matlab code for the chopped wing in `nonlinear_lifting_line/windex_fuselagetest`. The balance reference point is given in the `make_windexfusewing.m` script.

There is one important number to set based on this test in the Matlab script `make_windex.m`. The final statement

```
% fuselage drag in the form S*CD

  acg.ffsl = 0.0*acg.Sref;
```

can be used to add a increment in the drag to model the influence of the fuselage. This will give a much more realistic drag of the complete aircraft.

You can also use the wind tunnel test data to estimate the influence of the fuselage on your neutral point estimate and the lateral aerodynamics.

Finally, make sure to compare you aerodynamics modeling with the data provided by Ridder [7].

Task 4: Unsteady aerodynamics data

Use your model to estimate the unsteady aerodynamic coefficients due to rates of rotation around all three axes (p,q,r). You can also get the coefficients dependent on the angle of side slip β as they will be useful later. You can limit your modeling to small angles in the lateral degrees of freedom and to small rates of rotation.

Rates of rotation are set by the `omega` array in the flight state. The definitions in the aerodynamics model is slightly different so make sure to define the rates of rotation carefully. Check the scripts `coefficients.m` and `flight_state.m` for details. A typical sequence can be:

1. Set or check flight condition (speed, α,β, omega etc)
2. Run pointsolve.m to get vortex sheet (fs.gamma)
3. Run coefficients.m to get nondimensional forces and moments

It is also useful to use the approximate values from Chapter 9.13 in [3] to check that definitions and sign are correct. Compare your computed unsteady data with those in [7].

Task 5: Equations of motion

Templates for flight simulation in the longitudinal degrees of freedom are now available in folder `flightsim` on canvas. The main files to consider are:

`main.m`	Sample main program for trim and simulation.
`make_fsim.m`	Define aircraft data.
`fplmod.m`	Definition of differential and output equations.
`lon_aero.m`	Interpolation of longitudinal data.
`simsub.m`	Wrapper function for time integration of model.
`isaatm.m`	Atmosphere data, same as in previous tasks.

Additional files are available for a full 3D model to be used in later tasks.

The flight simulation model is implemented in standard flight mechanics notation and definitions as described in Etkin [5, 4] and Drela [3, Ch 9]. The main program trims the aircraft for level flight using elevator and throttle as control variables.

In order to use the model in a standard setting for solving initial value problems in Matlab, the control variables needs to be defined for the full time interval of the simulation. The controls are therefore set before the Matlab solver is invoked as illustrated in the main program and `simsub.m` subroutine.

Use the templates to complete the following subtasks:

1. Transfer your own aerodynamics database from previous tasks to the simulation model. This will require changes to `make_fsim.m`, `fplmod.m` and `lon_aero.m`. Use a fixed reference point for your aerodynamics database.

2. Implement transformation of the aerodynamics forces and moments to a center of gravity position that may be different from the aerodynamics reference position. Be careful in this step since the coordinate systems are different in the flight mechanics and aerodynamics modeling.

3. Modify the trim equations in the main program to trim the aircraft for gliding flight with throttle set to zero. Hint: use variables u, w, θ and elevator and functions for zero acceleration and constant airspeed.

4. Use your modified trim functionality to compute a new glide polar and compare to your previous glide polar. Small differences may show depending on your modeling both in previous tasks and how you implement the aerodynamics data in the simulation model.

5. Explore the use of the time integration function with ODE45 by changing control input and time interval.

Advanced topics*

Implement the additional data for lateral aerodynamics, flap and aileron control in the simulation model. Templates for this are available in the files `main3d.m` and `fplmod3d.m` which also define the full rigid body equations with 12 state variables. You can assume in your modeling that lateral aerodynamics states such as β, p and r are small in magnitude.

Task 6: Stability and control

Use the supplied main program for trimming the aircraft in the longitudinal degrees of freedom and perform the following set of analyses. Use your personal data for weight and balance and your choice of suitable position of the center of mass. For problem parameters not given, chose your own data and motivate your selection of parameters.

1. Trace the longitudinal stability eigenvalues as a function of airspeed in a root locus plot. Start at the lowest speed for which you can safely trim and fly the aircraft and continue up to your maximum safe speed to fly.
2. Compute period and time to double or time to half for the longitudinal stability modes as a function of airspeed.
3. Explore what happens to the above computed information if you shift the center of mass forward and reward.
4. Assess if the dynamic stability analysis will influence your recommended range of allowed center of mass positions.
5. Derive a linearized model for the longitudinal degrees of freedom with elevator and throttle as control variables. Select a few flight conditions of interest to you for analysis.
6. Use the linear model to explore the response to control input in both elevator and throttle using Matlab/Octave built-in functions for initial, step, impulse and frequency response (bode).
7. In particular make sure to investigate what a step input in the controls does to the flight state in terms of speed and flightpath angle. You will need to add an output variable for the flightpath angle.
8. Explore the response to control input and assess your choice of stability margins again.

It is also of interest to compare your stability analysis with simplified expression for the various modes. Such expressions are derived and presented in both Etkin [5, 4] and Drela [3, Ch 9]. The simplified expressions can in some cases give insight into what aircraft design parameters influence dynamic stability.

Advanced topics*

Derive linearized equations of motion also for the lateral degrees of freedom. You can trim the aircraft using the nonlinear longitudinal model since it is sufficient to model only small perturbations in the lateral degrees of freedom.

The perform the same type analysis as listed for the longitudinal degrees of freedom and explore the flying qualities of your aircraft model.

Task 7: Simulation of flight

The purpose with the this task is to use the developed flight simulation model to perform a few simulation tasks to investigate the performance of the aircraft.

Stall

Trim the aircraft in a glide at 1000 meters altitude at 30 m/s true airspeed. Perform a stall maneuver and make sure to reach at least 10 degrees angle of attack before recovering. Estimate the loss of altitude for the whole maneuver when back at 30 m/s true airspeed.

Looping

Trim the aircraft in a glide at 1000 meters altitude at an appropriate airspeed, for example 75 m/s true airspeed. Perform a proper looping without stalling the aircraft or reaching an excessive load factor. For this maneuver, do not exceed 6 g. Make sure to exit at the same altitude as when entering the looping and that the true air speed after the looping is not close to the stall speed.

Advanced topics*

Design a control system that maintains airspeed using throttle as input parameter. Design your control system for a suitable airspeed and tune the control system parameters for good response. Investigate how the control system performs at other trim conditions by linearizing the system at different altitude and speed settings. For how large range of altitude and speed is control system performance acceptable?

Acknowledgements

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